

Seminar: Image Segmentation Based on Visual Prompt

Group MNIST

VNUHCM - University of Science

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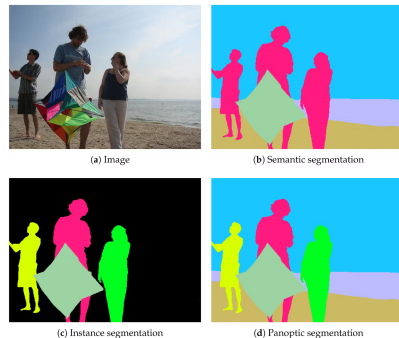
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General Background

Image segmentation is a fundamental task in computer vision that involves partitioning a digital image into multiple segments or sets of pixels, often corresponding to different objects or parts of objects. The goal is to simplify or change the representation of an image into something that is more meaningful and easier to analyze.



The history of this field can be broadly divided into three distinct eras.

- **Pre-Deep Learning Era (c. 1970s–2000s)**
- **Deep Learning Era (c. 2010s)**
- **Modern Research Directions**



Pre-Deep Learning Era (c. 1970s-2000s)

Segmentation methods were primarily based on handcrafted features and mathematical models. These techniques required significant domain knowledge and focused on pixel-level properties such as intensity, color, texture, and edges.



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Prominent methods from this period include:

- Canny Edge Detection
- K-Means Clustering
- Mean Shift Clustering



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These methods, while foundational, relied on heuristics and low-level features, often struggling with complex scenes and semantic understanding.



Deep Learning Era (c. 2010s)

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Key milestones include:

- Fully Convolutional Networks (FCN)
- U-Net
- Mask R-CNN



Current research continues to build upon these deep learning foundations, exploring new architectures and learning paradigms for greater accuracy, efficiency, and generalization.

- Transformer-Based Architectures
- Foundation Models
- Self-Supervised Learning
- Visual Prompt Model



The drive to advance image segmentation stems from both scientific inquiry and practical necessity.

Scientifically,

- Segmentation addresses a fundamental question in perception: while image classification tells us what is in an image, segmentation answers where it is, down to the pixel level; segmentation helps reveal how an image is organized by separating it into meaningful regions.
- Many methods in computer vision rely on segmented regions as inputs. Examples include object detection, recognition, tracking, 3D reconstruction, and medical image analysis. Better segmentation improves the performance of these tasks.
- Segmentation studies connect computational models with how humans interpret scenes. This benefits fields such as cognitive science and computational neuroscience.

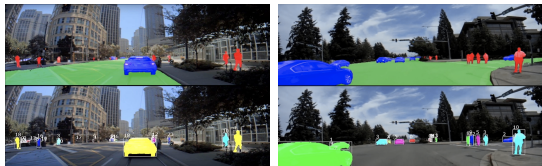


Practically, segmentation is an enabling technology for a multitude of real-world applications.

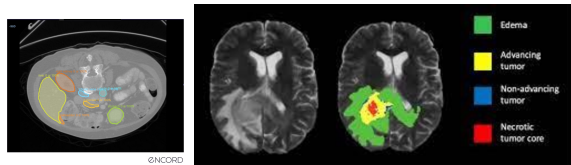


Motivation

Autonomous Driving



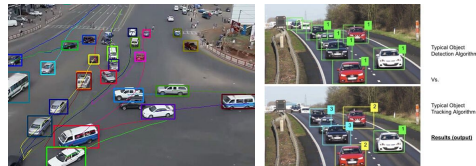
Medical Imaging



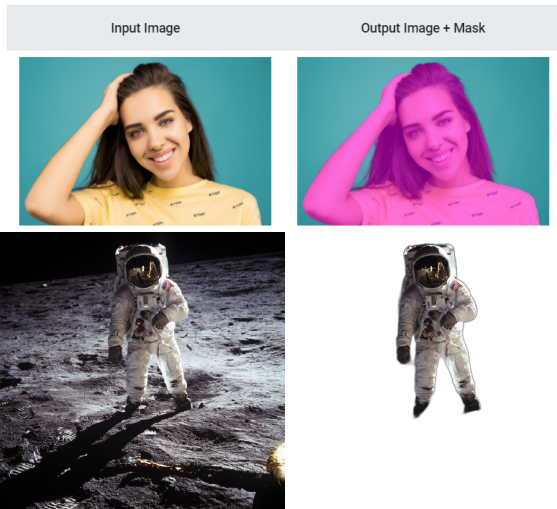
Surveillance



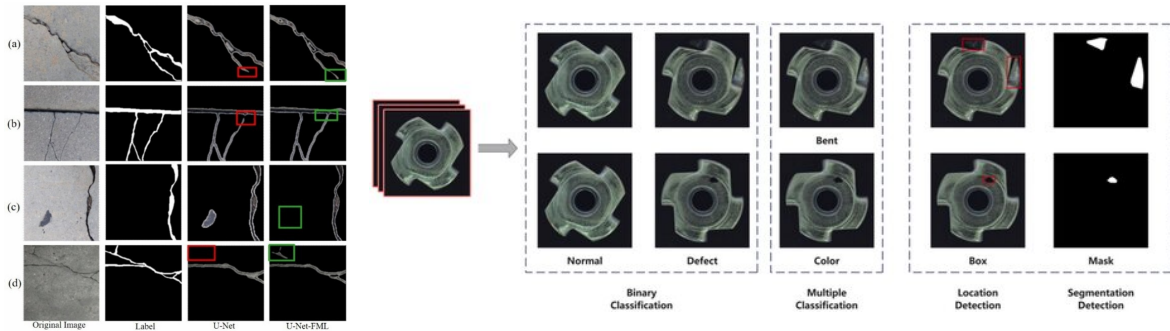
Object Tracking



Digital Photograph



Industrial Inspection



Segment Anything Model (SAM)

“In this work, our goal is to build a foundation model for image segmentation.” [?]

With this model, we aim to solve a range of downstream segmentation problems on new data distributions using prompt engineering.



Problem Statement

Image segmentation is the task of grouping pixels that belong to a common semantic object and separating them from other objects and the background.

The core principle of segmentation is guided by two criteria: pixels within the same group should be as similar as possible, while pixels belonging to different groups should be as distinct as possible.

In essence, the task simulates the human cognitive ability to transform raw, unstructured visual data into a structured representation of distinct entities.



Problem Statement

Let $\Omega \subset \mathbb{R}^2$ be the image domain and $I : \Omega \rightarrow \mathbb{R}^k$ the image.

A segmentation is a partition

$$\Omega = \cup_{i=1}^K \Omega_i \quad \Omega_i \cap \Omega_j = \emptyset \text{ for } i \neq j$$

such that pixels within each region share similar properties according to some criterion.



Problem Statement

For a traditional image segmentation method, the task can be described by its inputs and outputs.

- **Input:** An image together with a specified number of classes. Depending on the method, additional choices such as a similarity measure or clustering criterion may be required.
- **Output:** A segmentation result in which each pixel is assigned to one of the given classes. The output is typically represented as a label map where each pixel is assigned to a class of the same spatial size as the input image.



Problem Statement

For a modern promptable segmentation system like **SAM**, the problem can be formally defined by its inputs and outputs.

- **Input:** An image and a "visual prompt" that specifies the object of interest. This prompt can take various forms, such as a point (or multiple points) on the object, a bounding box enclosing it, or even a rough mask.
- **Output:** One or more segmentation masks for the object(s) indicated by the prompt, along with a confidence score for each mask. The ability to output multiple masks is crucial for handling ambiguity.

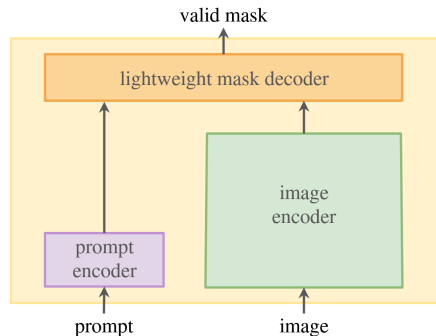


Figure 1: Model: Segment Anything Model (SAM)

"Our principal contributions are a new task (promptable segmentation), foundational model for that new task (SAM), and dataset (SA-1B)"

A key contribution is the SA-1B dataset, which remains the largest segmentation dataset to date. Its key features include:

- **Scale:** It contains over 11 million high-resolution images and 1.1 billion high-quality segmentation masks.
- **Privacy-First:** All images are privacy-respecting, with automated blurring of faces and license plates.
- **Diversity:** The dataset is geographically diverse, with a significant number of images from countries around the world, as illustrated in 2. This helps mitigate geographical bias often present in large-scale datasets.



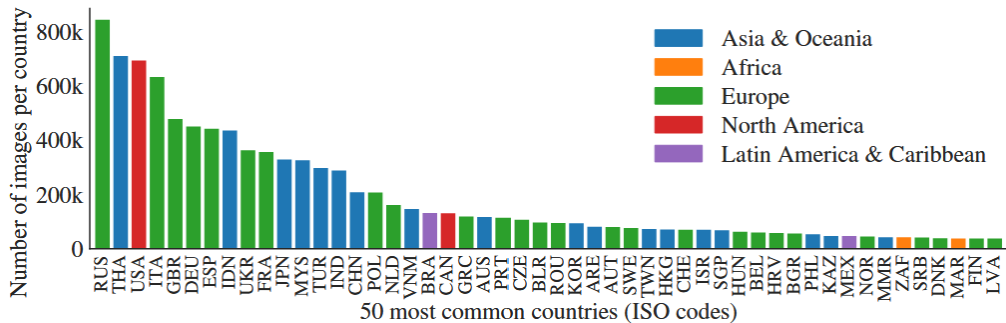


Figure 2: Geographical Diversity of the Dataset

The contribution of this report is to provide a comprehensive survey and structured analysis of the field of image segmentation. We synthesize the historical evolution, explain the motivations driving modern research, and offer a detailed examination of the methodology behind foundation models like SAM, contextualizing their impact and outlining future directions.

In traditional methods, there are different techniques for image segmentation like threshold-based, edge-based, cluster-based.



Traditional Methods

Methods	Mechanics
K-means clustering	Iteratively assigns each pixel to the nearest cluster center and updates centers using the mean of assigned pixels.
Mean shift clustering	Moves each pixel toward the densest region of the feature space using kernel density estimation until convergence.
Binary thresholding	Compares each pixel intensity with a chosen threshold and assigns it to foreground or background.



Traditional Methods

Methods	Methodology	
	Stage 1: Input preparation	Stage 2: Parameter or criterion selection
Binary thresholding	Uses pixel intensity values directly	Choose threshold value manually or automatically
K-means clustering	Uses pixel features such as intensity, color or coordinates	Select the number of clusters K
Mean shift clustering	Maps pixels into a feature space and prepares a kernel window	Choose kernel bandwidth defining neighborhood size



Methods	Methodology	
	Stage 3: Decision rule or region formation	Stage 4: Final labeling
Binary thresholding	Compare intensity with threshold and classify each pixel	Produce a binary mask with two classes
K-means clustering	Assign pixel to the nearest cluster center and refine iteratively	Output cluster label for each pixel
Mean shift clustering	Shift pixel toward density peak and group by convergence mode	Assign each pixel to the mode it reaches

Traditional Methods

Methods	Contribution
Binary thresholding	Provided an efficient method for separating foreground and background and remains widely used in preprocessing and simple segmentation tasks.
K-means clustering	Introduced a simple and widely used baseline that works across many image types and served as a foundation for later clustering methods.
Mean shift clustering	Demonstrated a nonparametric density based approach that does not require specifying the number of clusters and influenced modern segmentation research.



Traditional Methods

Methods	Limitations
Binary thresholding	Works only when intensity separation is clear. Sensitive to lighting changes. Cannot handle multiple classes without extension.
K-means clustering	Requires the number of classes beforehand. Sensitive to initialization. Converges to local minima. Performs poorly with non convex clusters.
Mean shift clustering	High computational cost for large images. Sensitive to kernel bandwidth. May over segment when parameters are not tuned.



Binary Threshold

Binary thresholding is used to separate the foreground (region of interest) from the background based on pixel intensity values. It is particularly effective when the objects and background have distinct intensity distributions.



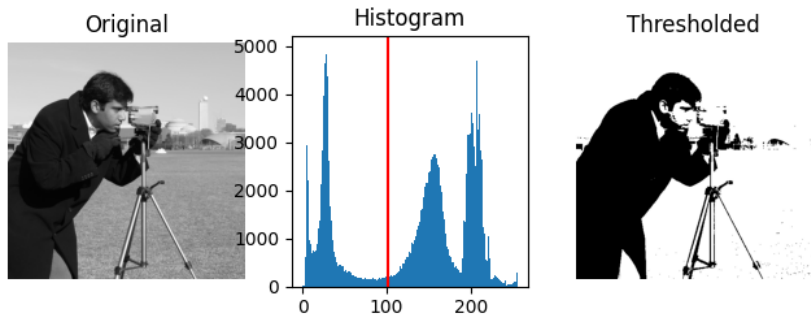
Original Image



Thresholded and segmented Image

Binary Threshold

Each pixel in the grayscale image is compared against a threshold value T . Pixels with intensity greater than or equal to T are classified as foreground, while the others are classified as background. This process converts a grayscale image into a binary one.



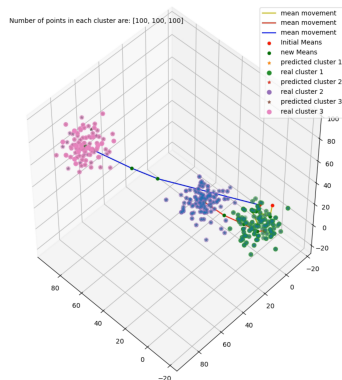
K-Means Clustering

K-Means is an unsupervised clustering algorithm used to segment regions of interest from the background [Nameirakpam et al., 2015]. It was first introduced by Hartigan in 1975, and its goal is to partition M points in an N -dimensional space into K clusters such that the within-cluster sum of squares is minimized [Hartigan and Wong, 2018].



K-Means Clustering

The algorithm iteratively assigns each pixel to the nearest cluster centroid and updates centroids based on current assignments. K-Means algorithm seeks a local optimum rather than a global one. Data points are divided into groups by assigning each of them to the nearest centroid. The goal is to find a solution such that moving any point from one cluster to another does not reduce the within-cluster sum of squares [Hartigan and Wong, 2018].



Mean Shift Clustering

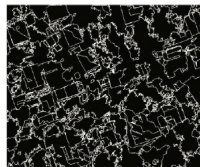
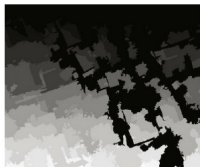
Mean Shift is a non-parametric clustering algorithm, which falls into unsupervised clustering category. Unlike K-Means, it does not require specifying the number of clusters in advance. Instead, it locates regions of high data density (modes) by iteratively shifting data points toward the nearest local maximum of a kernel density estimate [Comaniciu and Meer, 2002].



Original Satellite Imagery

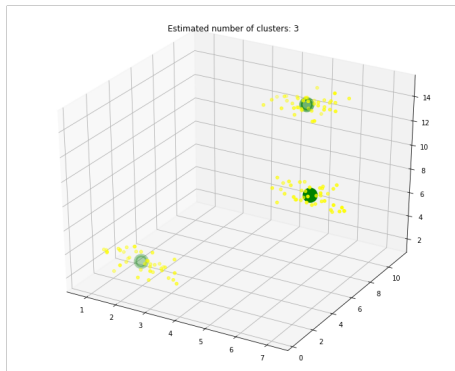


Mean Shift Clustered Imagery



Mean Shift Clustering

The algorithm treats data points as samples from an underlying probability density function and seeks its local maxima. Each pixel is iteratively moved toward regions of higher density, determined by a kernel function, until convergence. Pixels converging to the same mode belong to the same cluster.



Technique	Mechanics
Fully connected networks	Treat the image as a long vector and process it through dense layers with no spatial preservation.
U-Net	Uses an encoder to compress features and a decoder with skip connections to restore spatial detail for pixel level prediction.
Mask R-CNN	Extends Faster R CNN by adding a mask prediction branch that produces instance level segmentation for each detected object.

Technique	Methodology	
	Input preparation	Core mechanism
Fully connected networks	Flatten the image into a vector of pixel values	Apply dense layers to classify each pixel or entire image regions
U-Net	Feed the full image or patches as tensor input	Use encoder decoder with skip connections to learn detailed spatial features
Mask R-CNN	Use image as input along with region proposals from RPN	Combine feature backbone, region alignment and mask head for instance level masks

Technique	Methodology	
	Training process	Output generation
Fully connected networks	Train with backpropagation on flattened pixels using classification loss	Produce class probabilities for each pixel then reshape to segmentation map
U-Net	Train with pixel wise loss while leveraging skip connections to recover spatial detail	Generate dense segmentation map matching input resolution
Mask R-CNN	Train multi stage loss for bounding boxes, classes and per instance masks	Output object bounding boxes, class labels and a separate mask for each instance

Technique	Contribution
Fully connected networks	Provided early neural network baselines for segmentation but highlighted the need for spatially aware architectures.
U-Net	Became one of the most influential architectures in medical imaging and many pixel level tasks due to its skip connection strategy.
Mask R-CNN	Established a strong and widely adopted framework for instance segmentation by unifying detection and mask generation.

Technique	Limitations
Fully connected networks	Large number of parameters, loss of spatial information and poor performance on detailed segmentation.
U-Net	Performance depends on training data scale, can be memory heavy and is less suited for object level separation.
Mask R-CNN	Complex multi stage training pipeline and high computational cost for large images or many objects.

U-Net, proposed by Ronneberger et al. in 2015, features a symmetric encoder-decoder architecture with extensive skip connections [?]. Originally designed for biomedical image segmentation, U-Net achieves high accuracy even with limited training data.

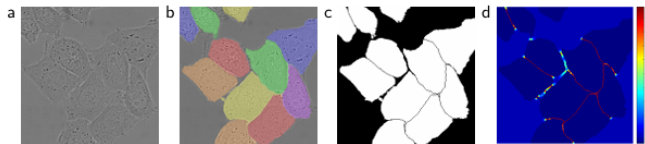
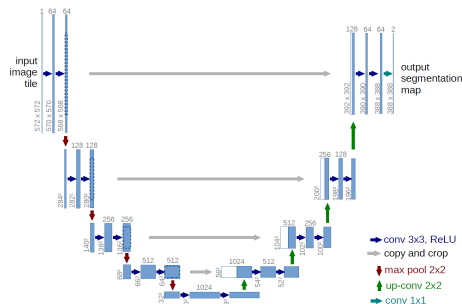


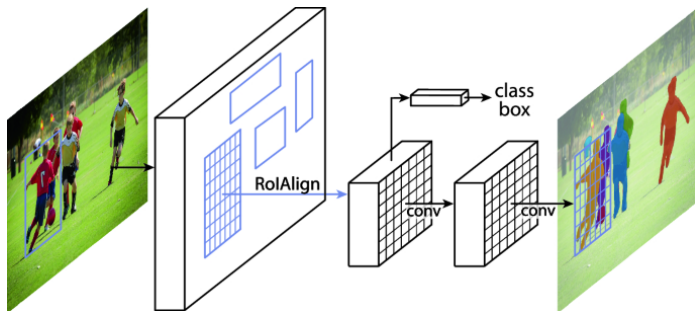
Fig. 3. HeLa cells on glass recorded with DIC (differential interference contrast) microscopy. (a) raw image. (b) overlay with ground truth segmentation. Different colors indicate different instances of the HeLa cells. (c) generated segmentation mask (white: foreground, black: background). (d) map with a pixel-wise loss weight to force the network to learn the border pixels.

FCN transforms classification networks (e.g., VGG, ResNet) into fully convolutional architectures by replacing dense layers with 1×1 convolutions. The network produces coarse feature maps through successive pooling, then upsamples them back to the original image resolution using transposed convolutions (deconvolution). Skip connections combine deep semantic features with shallow appearance features to recover spatial details lost during downsampling.



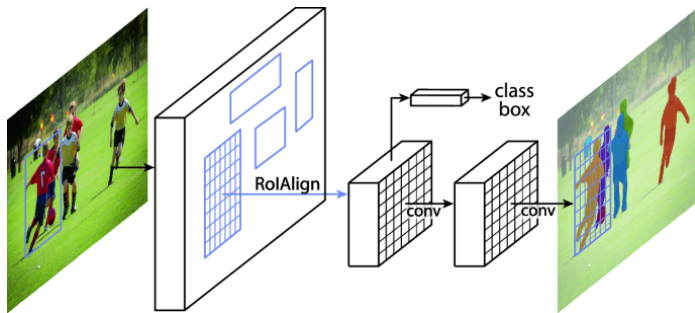
Mask R-CNN

Mask R-CNN, introduced by He et al. in 2017, extends Faster R-CNN for instance segmentation by adding a parallel branch for predicting segmentation masks [He et al., 2017]. This architecture excels at detecting and segmenting individual object instances.

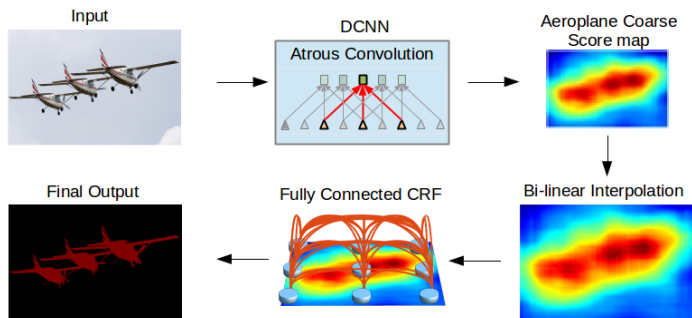


Mask R-CNN

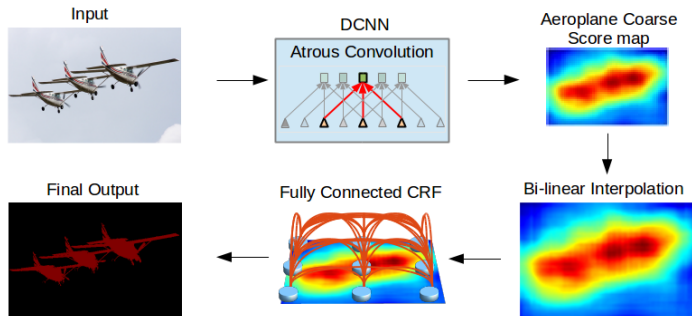
Mask R-CNN operates in two stages. First, a Region Proposal Network (RPN) generates candidate object bounding boxes. Second, for each region of interest (RoI), the network performs three parallel tasks: class prediction, bounding box regression, and mask prediction. The key innovation is RoIAlign, which preserves spatial alignment during RoI pooling by using bilinear interpolation, crucial for pixel-accurate mask prediction.



DeepLab, developed by Chen et al., employs atrous (dilated) convolution and Atrous Spatial Pyramid Pooling (ASPP) to capture multi-scale contextual information without reducing spatial resolution [?]. DeepLabv3+ further incorporates an encoder-decoder structure.

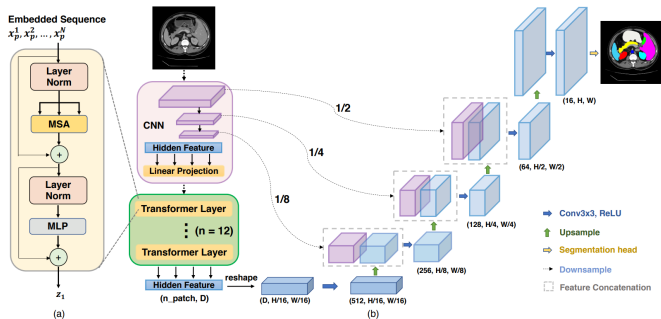


DeepLab uses atrous convolution to expand the receptive field without increasing parameters or reducing resolution. The atrous rate r controls the spacing between kernel weights. ASPP applies parallel atrous convolutions with different rates to capture objects at multiple scales. The encoder-decoder structure (in DeepLabv3+) first extracts features, then gradually recovers spatial details through the decoder with skip connections.



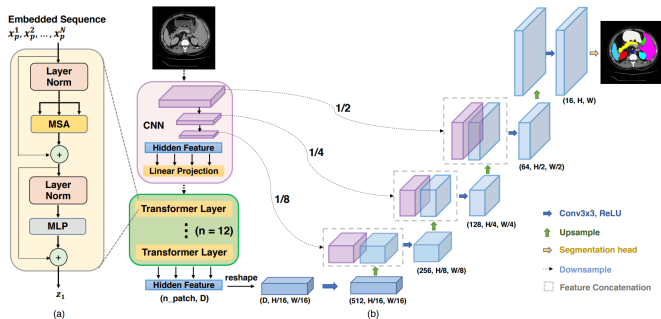
Transformer-Based Segmentation

Vision Transformers (ViTs) adapt the self-attention mechanism from NLP to computer vision. TransUNet, introduced by Chen et al. in 2021, combines Transformers with U-Net architecture for medical image segmentation [Chen et al., 2021].



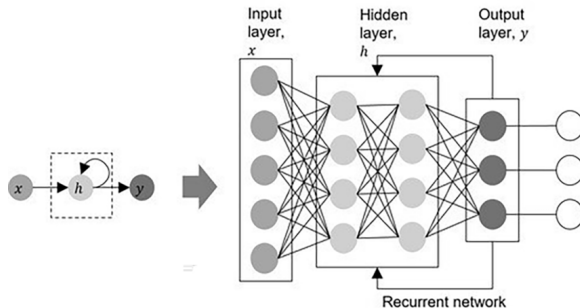
Transformer-Based Segmentation

TransUNet uses a CNN to extract initial features, then patches the feature map and processes it through Transformer encoder layers. The self-attention mechanism captures long-range dependencies between image patches, overcoming the limited receptive field of CNNs. The Transformer output is reshaped and fed into a CNN decoder similar to U-Net, with skip connections from the CNN encoder for recovering spatial details.



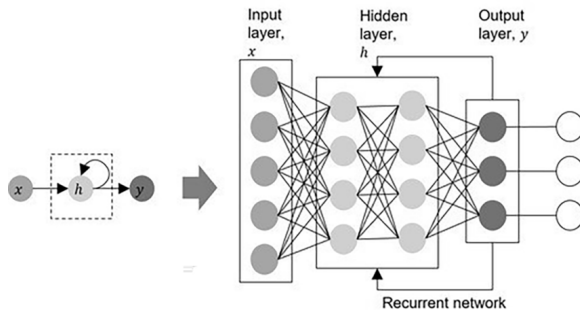
Recurrent Neural Networks for Segmentation

RNNs, particularly Long Short-Term Memory (LSTM) networks, have been applied to image segmentation to model spatial dependencies and context [?]. ReNet and related architectures process images sequentially in horizontal and vertical sweeps.



Recurrent Neural Networks for Segmentation

RNN-based segmentation treats the image as a sequence, processing it row-by-row or with overlapping patches. Four RNN layers sweep the image in different directions (left-to-right, right-to-left, top-to-bottom, bottom-to-top) to capture context from all spatial neighbors. The hidden states from all four directions are concatenated to form a rich contextual representation for each pixel. Multiple ReNet layers can be stacked to increase the receptive field.



Technique	Mechanics
GrabCut	Models foreground and background with Gaussian Mixture Models.
FocalClick	Dual branch design: global branch for full image understanding, focal branch that zooms into the region indicated by clicks.
SAM	Uses a large image encoder based on Vision Transformers and a lightweight prompt encoder for points, boxes, or masks.



Visual Prompt Methods

Technique	Methodology	
	Prompt Acquisition and Encoding	Feature Representation and Context Modeling
GrabCut	Bounding box serves as the prompt and is encoded by assigning initial labels to foreground and background regions	Image representation is formed by GMM based color models and pair-wise smoothness terms
FocalClick	Clicking points are encoded as foreground or background click maps that serve as spatial guidance	Contextual representation is obtained using deep features from global and focal branches
SAM	Prompts such as points, boxes, or masks are encoded into prompt tokens via a dedicated prompt encoder	Contextual representation is obtained from a Vision Transformer that produces a high dimensional embedding of the input image



Technique	Methodology	
	Guided Inference	Mask Generation
GrabCut	Guided inference occurs through graph cut energy minimization	Mask generation corresponds to the binary labeling produced by the optimized cut
FocalClick	Guided inference is performed by a learned mapping that conditions feature integration on the encoded prompts	Mask generation is achieved by fusing outputs from global and focal modules to produce a refined prediction
SAM	Guided inference is conducted by a mask decoder that integrates prompt embeddings with the image embedding using attention mechanisms	Mask generation consists of producing multiple candidate masks with associated confidence scores in a single forward pass



Technique	Contribution
GrabCut	Introduced iterative GMM refinement integrated with graph cuts. Became a foundational technique in classical computer vision.
FocalClick	Introduced coarse to fine refinement through focal regions. Better handling of small objects and thin structures than older deep methods.
SAM	First truly general purpose segmentation foundation model. Enables interactive segmentation, automatic segmentation, and box guided segmentation in one model.

Technique	Limitations
GrabCut	Struggles with fine boundaries such as hair or transparent objects. Sensitive to the initial bounding box.
FocalClick	Does not generalize as well to unseen image types compared to foundation models. Performance depends on click placement quality.
SAM	High computational cost during preprocessing due to large encoder. Not ideal for pixel perfect tasks such as thin structures or high precision medical boundaries without fine tuning. Mask quality depends on prompt placement.

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